Contents

Chapter 4 — The Concept of a Random Variable

4-1 Introduction

Definitions

- **Random Variable**: A random variable $x(\zeta)$ is a process of assigning a real number to every outcome ζ of an experiment.
  - Its domain S is the set of all experimental outcomes, and its range is a set of numbers on the real axis.
  - To qualify as a random variable, it must satisfy two mathematical conditions:
    1. The set $\{x \leq x\}$ must be an event for every real number x.
    2. The probabilities of the events $\{x = \infty\}$ and $\{x = -\infty\}$ must equal 0: $P\{x = \infty\} = 0$ and $P\{x = -\infty\} = 0$.
- **Complex Random Variable**: Defined as $z = x + jy$, where x and y are real random variables.

4-2 Distribution and Density Functions

Definitions

- **Cumulative Distribution Function (CDF)**: The distribution function of the random variable x is defined as $F_x(x) = P\{x \leq x\}$, for every x from $-\infty$ to ∞.
- **Probability Density Function (PDF)**: The derivative of the probability distribution function: $f_x(x) = \frac{dF_x(x)}{dx}$.

[Image of cumulative distribution function]

Corollaries and Properties

Properties of the Distribution Function $F(x)$: 1. Limits: $F(+\infty) = 1$ and $F(-\infty) = 0$. 2. Non-decreasing: If $x_1 < x_2$, then $F(x_1) \leq F(x_2)$. 3. Complement probability: $P\{x > x\} = 1 - F(x)$. 4. Right-continuous: $F(x^+) = F(x)$. 5. Interval probability: $P\{x_1 < x \leq x_2\} = F(x_2) - F(x_1)$. 6. Point probability: $P\{x = x\} = F(x) - F(x^-)$.

Properties of the Density Function $f(x)$: * Non-negativity: $f_x(x) \geq 0$ for all x. * Unit area: $\int_{-\infty}^{\infty} f_x(x)dx = 1$. * For continuous-type random variables, the probability in a small interval is approximated by: $P\{x \leq \mathbf{x} \leq x + \Delta x\} \approx f(x)\Delta x$.

Percentiles

- u **Percentile**: The smallest number x_u such that $u = P\{x \le x_u\} = F(x_u)$. The **median** is defined as the 0.5 percentile.

4-3 Specific Random Variables

Existence Theorem

- Given a non-negative function $f(x)$ whose integral is 1, or a right-continuous, monotonically increasing function $F(x)$ from 0 to 1, we can construct an experiment and a random variable x with distribution $F(x)$ or density $f(x)$.

Examples

Common Continuous-Type Random Variables: 1. **Normal (Gaussian) Distribution**: $x \sim N(\mu, \sigma^2)$. The density function is $f_x(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$.

[Image of normal distribution probability density function]

2. **Exponential Distribution**: $f_x(x) = \lambda e^{-\lambda x}$ for $x \ge 0$, and 0 otherwise. It exhibits the memoryless property.

[Image of exponential distribution probability density function]

3. **Uniform Distribution**: $x \sim U(a, b)$. $f_x(x) = \frac{1}{b-a}$ in the interval (a, b), and 0 otherwise.
4. Other notable continuous distributions include Chi-square, Rayleigh, and Beta distributions.

Common Discrete-Type Random Variables: 1. **Bernoulli Distribution**: Takes the values 1 (with probability of success p) and 0 (with probability of failure $q = 1-p$). 2. **Binomial Distribution**: Represents the number of successes in n independent Bernoulli trials. $P\{y = k\} = \binom{n}{k} p^k q^{n-k}$. 3. **Poisson Distribution**: $P\{x = k\} = \frac{e^{-\lambda}\lambda^k}{k!}$, often used to model the number of occurrences of a rare event in a large number of trials. 4. **Geometric Distribution**: Denotes the number of trials needed to realize the first success. It also possesses the memoryless property.

4-4 Conditional Distributions

Definitions

- **Conditional Distribution Function**: Assuming an event M where $P(M) \neq 0$, the conditional distribution function of x is defined as:

$$F(x|M) = P\{x \le x|M\} = \frac{P\{x \le x, M\}}{P(M)}$$

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- **Conditional Density Function**: The derivative of the conditional distribution function: $f(x|M) = \frac{dF(x|M)}{dx}$. This function is non-negative and its total area equals 1.

Properties

- Similar to standard distributions, $F(\infty|M) = 1$ and $F(-\infty|M) = 0$.
- Conditional interval probability: $P\{x_1 < x \le x_2|M\} = F(x_2|M) - F(x_1|M)$.

Examples of Conditional Densities

- **Conditioned on a one-sided interval** $M = \{x \le a\}$:
  - $F(x|x \le a) = \frac{F(x)}{F(a)}$ for $x < a$, and equals 1 for $x \ge a$.
  - The corresponding conditional density is $\frac{f(x)}{F(a)}$ for $x < a$, and 0 otherwise.
- **Conditioned on a bounded interval** $M = \{b < x \le a\}$:
  - The conditional density is $\frac{f(x)}{F(a)-F(b)}$ for $b \le x < a$, and 0 otherwise. This is known as a truncated distribution.

Theorems: Total Probability & Bayes'

- **Continuous Total Probability Theorem**: If the events $A_1, \dots, A_n$ form a partition of the space S, the marginal density is:

$$f(x) = \sum_i f(x|A_i)P(A_i)$$

  . Its continuous integral form is $\int P(A|x = \mathbf{x})f(x)dx = P(A)$.
- **Continuous Bayes' Theorem**:

$$f(x|A) = \frac{P(A|x = \mathbf{x})f(x)}{P(A)} = \frac{P(A|x = \mathbf{x})f(x)}{\int P(A|x = \mathbf{x})f(x)dx}$$

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